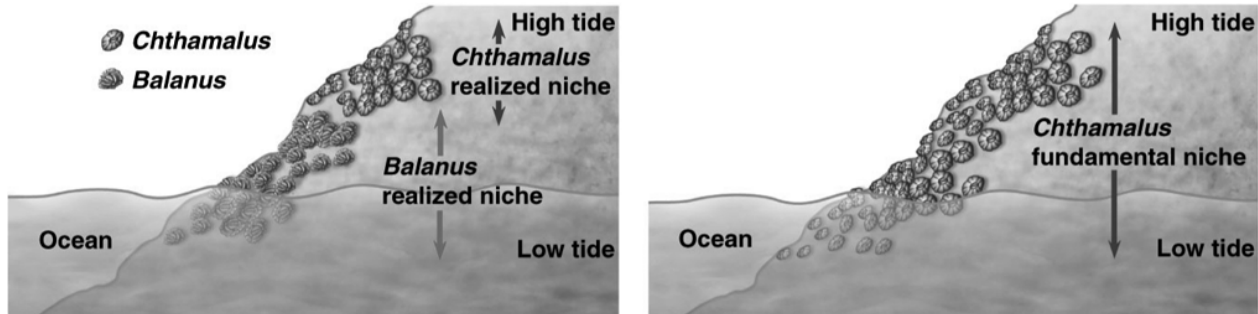


Each question is worth 2 points except where noted. Keep on truckin'...

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

1. In the experiment shown in the figure below, *Balanus* was removed from the habitat shown on the left. Which of the following statements is a valid conclusion of this experiment?



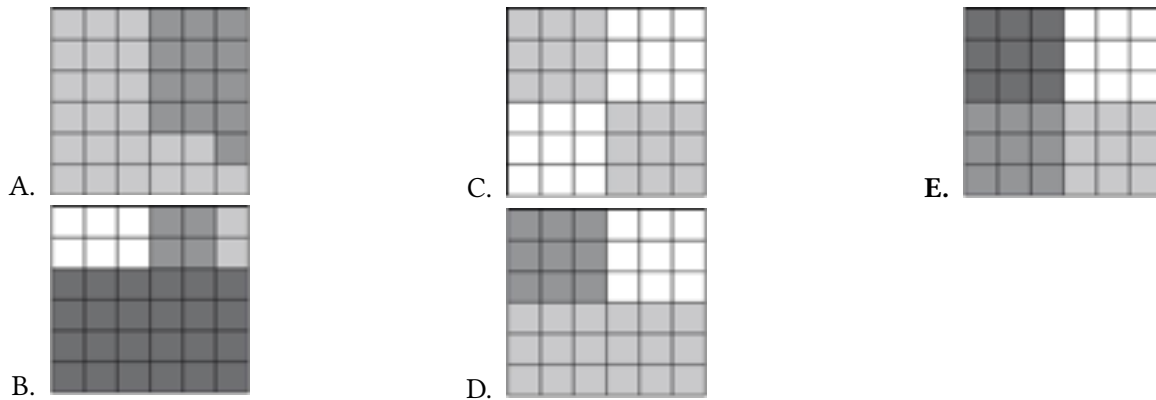
- A. **The removal of *Balanus* shows that the realized niche of *Chthamalus* is smaller than its fundamental niche.**
- B. If *Chthamalus* were removed, *Balanus*'s fundamental niche would become larger.
- C. These two species of barnacle do not show competitive exclusion.
- D. *Balanus* can survive only in the lower intertidal zone because it is unable to resist dehydration.
- E. *Balanus* is inferior to *Chthamalus* in competing for space on rocks lower in the intertidal zone.
2. The dispersion pattern for individuals in across the entire metapopulation shown in question 33 is most likely
- A. even.
- B. random.
- C. **clumped.**
- D. gene flow.
- E. uniform.
3. The "role" or "job" of a species in a community, represented by all of the resources needed by the species, is called
- A. a predator or prey.
- B. a predator.
- C. **the niche.**
- D. the resource.
- E. a producer or consumer.
4. Which of the following is most likely to contribute to density-dependent regulation of populations?
- A. Earthquakes.
- B. Fires.
- C. Floods.
- D. The removal of toxic waste by decomposers.
- E. **Intraspecific competition for nutrients.**

5. Given the following values and assuming exponential population growth, what will be the total population size after one generation has elapsed? Round to nearest whole individual.

$$b = 0.09 \quad N = 18,657$$

$$d = 0.12 \quad K = 15,000$$

- A. 3,657
 B. 560
 C. 18,097
 D. 15,000
 E. 19,217
6. Which block in the figure below is most consistent intermediate levels of disturbance in an ecosystem? (Different shades, including white, represent different species. The small squares indicate relative abundance of individuals present.)



7. Color that helps hide an animal in its environment is called
- A. aposomatic coloration
 B. hideandseek coloration.
 C. **cryptic coloration.**
 D. environmental coloration.
 E. warning coloration.
 F. mimicry.
8. Several species of warblers (a group of small colorful birds) in North America can live in the same tree only because they
- A. out-competed other species of birds, such as sparrows.
 B. can fly to different heights within the tree.
 C. eat different foods within the tree.
 D. **partitioned resources to eliminate competition.**
 E. created different types of nests within the tree.

9. Which of the following is *not* directly concerned with community ecology?
- A. predation
 - B. intraspecific competition**
 - C. herbivory
 - D. resource partitioning
 - E. symbiosis
10. Cowbirds lay their eggs in the nests of other birds like cardinals. The cardinals raise the cowbird chicks as their own but the cowbird chicks usually kill the cardinal chicks of the other bird. This is most likely an example of
- A. symbiosis.
 - B. mutualism.
 - C. parasitism.**
 - D. competition.
 - E. resource partitioning.
11. Competitive exclusion means that
- A. two species will partition resources to eliminate competition.
 - B. a species will have a smaller realized niche than a fundamental niche.
 - C. one species will always out compete another species that occupies the same niche.**
 - D. one individual will always out compete another individual in a population.
 - E. one population will always exclude another population of the same species in a community.

Circle True or False. Read each statement completely before answering.

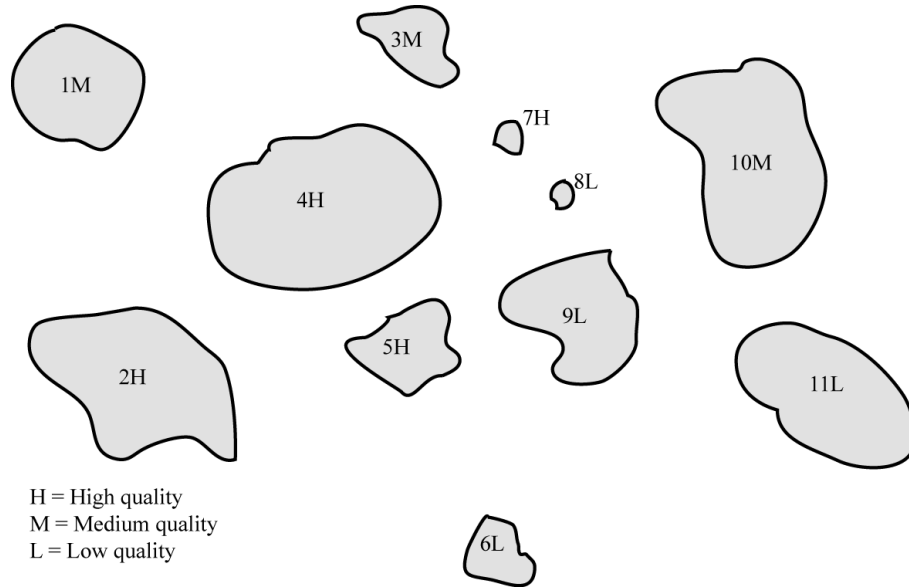
- 12. **TRUE** False The intrinsic rate of population growth does not include immigration or emigration.
 - 13. **TRUE** False Mutualism can be thought of as two species that ecologically exploit each other over the long term.
 - 14. **TRUE** False The brightly colored pattern of poison dart frogs is an example of aposomatic coloration.
 - 15. **TRUE** False The carrying capacity of a habitat may change as environmental conditions change.
 - 16. **TRUE** False Population is the lowest hierarchical level for the biosphere category.
 - 17. True **FALSE** Density-independent forces are most likely to affect population size when N is less than K .
 - 18. **TRUE** False A population may grow smaller if death rate is greater than immigration rate.
 - 19. True **FALSE** Most organisms grow under idealized conditions because they are adapted to their environment.
 - 20. True **FALSE** Evenness measures the distribution of relative abundance across all species in a population.
-

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Aposomatic	Exponential growth	Mimicry
Commensalism	Fundamental niche	Mutualism
Community	Habitat fragmentation	Niche compression
Corridor	Immigration	Parasitism
Cryptic coloration	Interspecific competition	Population
Diversity	Intraspecific competition	Realized niche
Ecosystem	Intrinsic growth rate	Resource Partitioning
Emigration	Keystone species	Richness
Evenness	Logistic growth	Symbiosis

21. **Realized niche** Based on the resources actually used by a species.
22. **Corridor** Can reduce local extinction by connecting sources and sinks.
23. **Emigration** Individuals leave a population.
24. **Parasitism** One species get ecological benefit over the long term, the other species is harmed.
25. **Richness** The total number of species present in the community.
26. **Interspecific Competition** Occurs when individuals of different species require the same limited resource.
27. **Resource Partitioning** Species divide shared resource, increasing diversity.
28. **Logistic Growth** Occurs when a population has a carrying capacity.
29. **Ecosystem** Interaction between organisms with each other and their physical environment.
30. **Mimicry** A tasty species looks like a poisonous species.
31. **Keystone Species** A species that greatly increases community diversity.
32. (2 points) Write the name of your favorite band or musical artist.

33. (15 points) Consider the following metapopulation. Identify *one* subpopulation that is *most likely* to have the highest extinction rate and lowest recolonization rate. Justify your choice by explaining how patch size, distance and habitat quality interact to affect extinction and recolonization in your chosen patch. There may be more than one correct choice but explain for *only* one subpopulation.



34. (6 points) You performed a t -test to compare a variable between two populations. The sample size from *each* population was 5 individuals. You calculated $t = 2.27$. Using $\alpha = 0.05$, do you accept or reject the null hypothesis? Explain how you used the table at right to decide, including specific numbers.

Table 1. Critical values of t .

df	α			
	0.10	0.05	0.02	0.01
1	6.31	12.71	31.82	63.66
2	2.92	4.30	6.96	9.92
3	2.35	3.18	4.54	5.84
4	2.13	2.78	3.75	4.60
5	2.02	2.57	3.36	4.03
6	1.94	2.45	3.14	3.71
7	1.89	2.36	3.00	3.50
8	1.86	2.31	2.90	3.36
9	1.83	2.26	2.82	3.25
10	1.81	2.23	2.76	3.17
11	1.80	2.20	2.72	3.11
12	1.78	2.18	2.68	3.06

Accept null hypothesis. Degrees of freedom is 8, so critical $t = 2.31$. Calculated $t = 2.27$, which is lower than the table value. If the calculated value is lower than the tabled value, then you accept (or fail to reject) the null hypothesis.

Here are four numbers: 37, 33, 53, 69.

35. (9 points) Calculate the mean and standard deviation for the numbers listed above. **Show your work for all steps** or you will not get any pudding. Round your final answers to two digits after the decimal. 2 points per blank plus 5 points for your work.

$$s = \sqrt{\frac{\sum (Y - \bar{Y})^2}{n-1}}$$

$$\bar{Y} = \mathbf{48.0}$$

$$s = \mathbf{16.45}$$

36. (6 points) Calculate the Shannon diversity index for the same four numbers above. You may use the $\ln$ or $\log$ key on your calculator. Show your work and round to two digits after the decimal for your final answer. 2 points per blank plus 4 points for your work.

$$H' = -\sum p_i \ln p_i$$

$$H' = 1.34 \text{ or } 0.58$$

Give them 2 points for each correct answer, even if they do not show their work.

